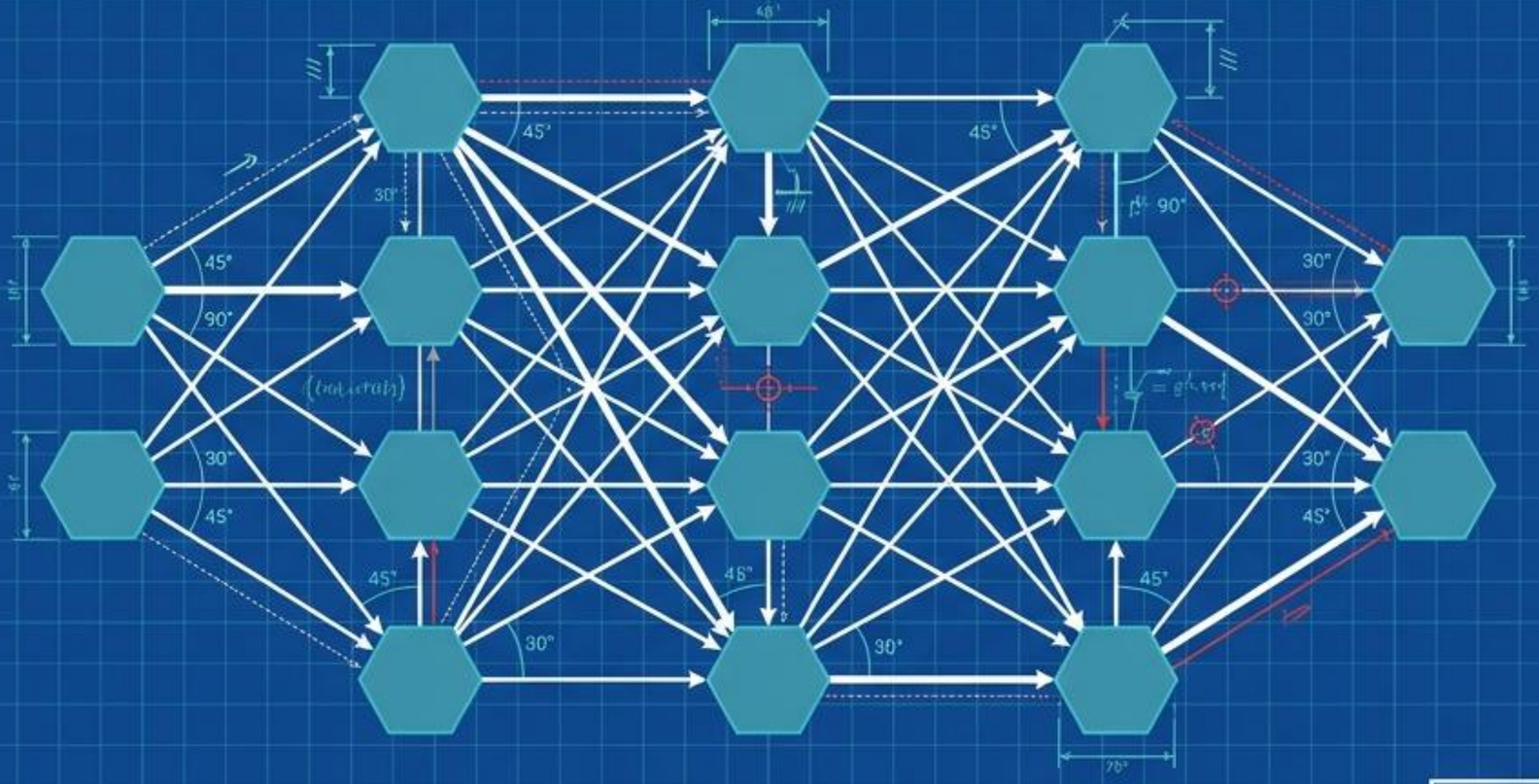
PAC LEARNING

The Theoretical Foundations of Machine Learning

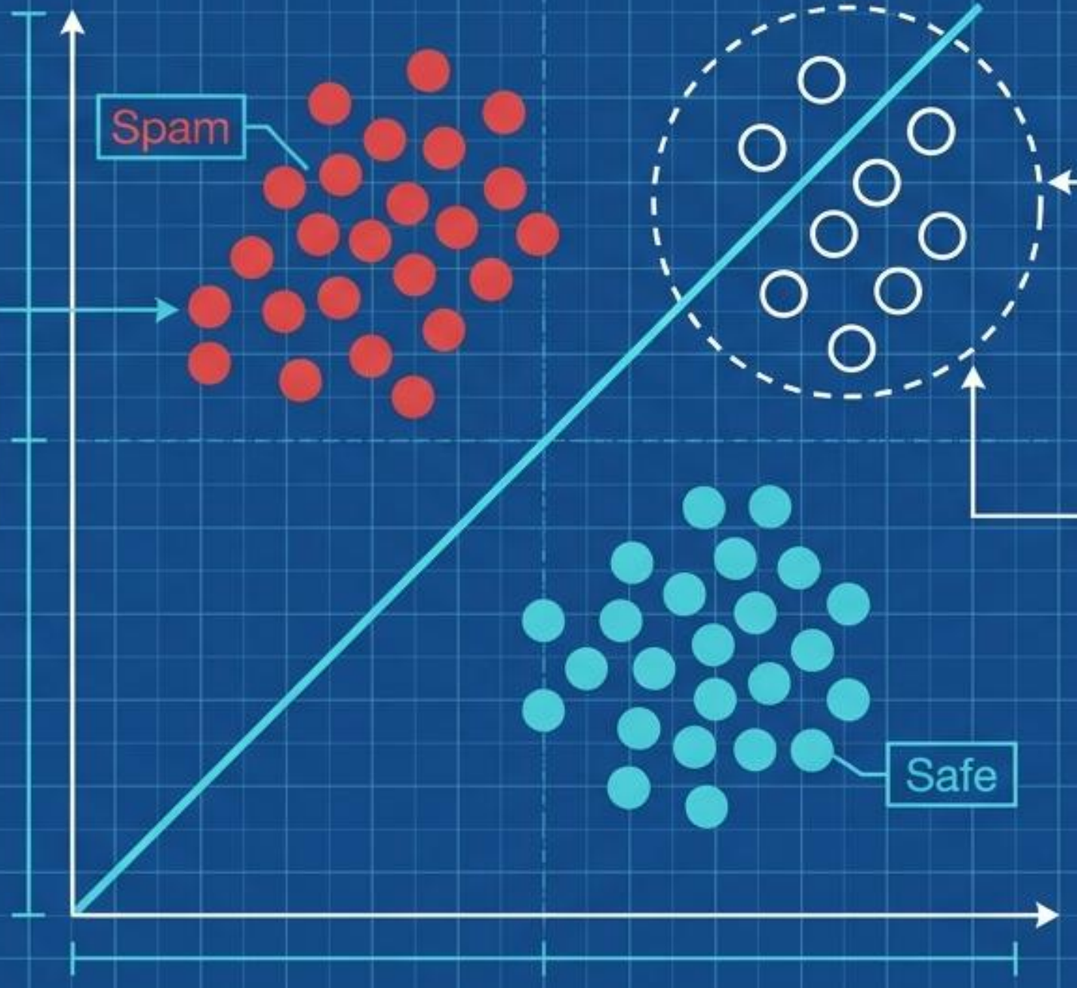


B.Tech CSE Advanced Modules

Blueprint v1.0

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The Anxiety of Generalization

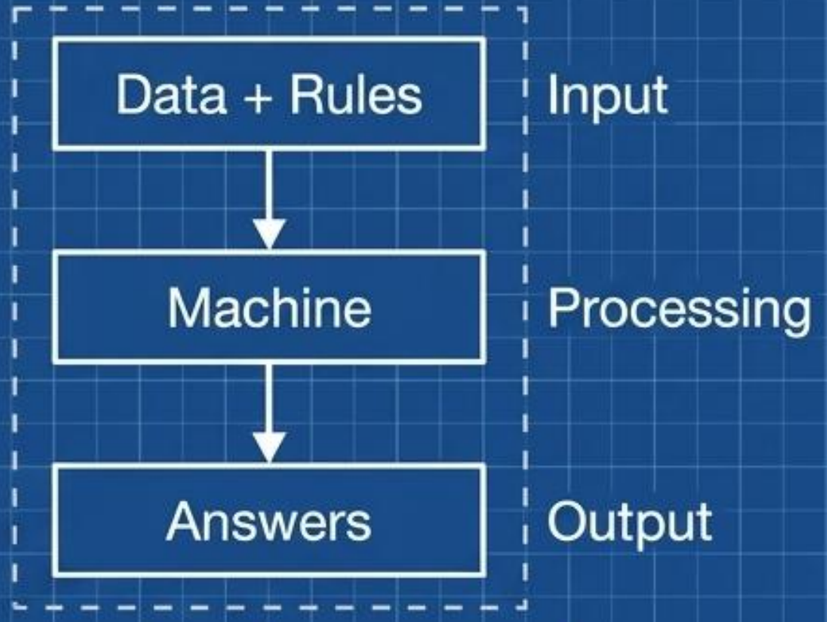


A Google Gmail spam classifier achieves 99% accuracy on yesterday's data.

The Motivating Problem: How do we *mathematically guarantee* it will classify tomorrow's unseen emails correctly?

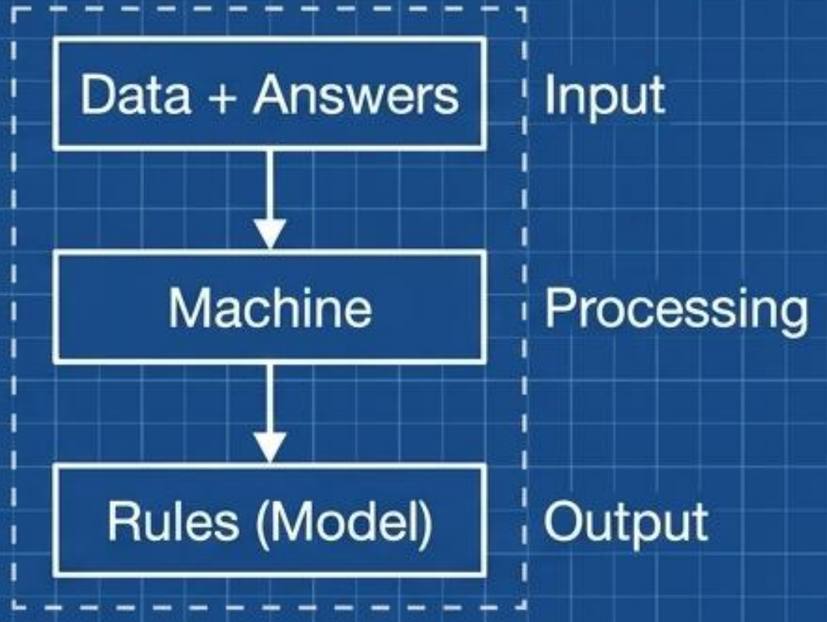
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Traditional Programming



Correctness is guaranteed by the programmer's logic.

Machine Learning



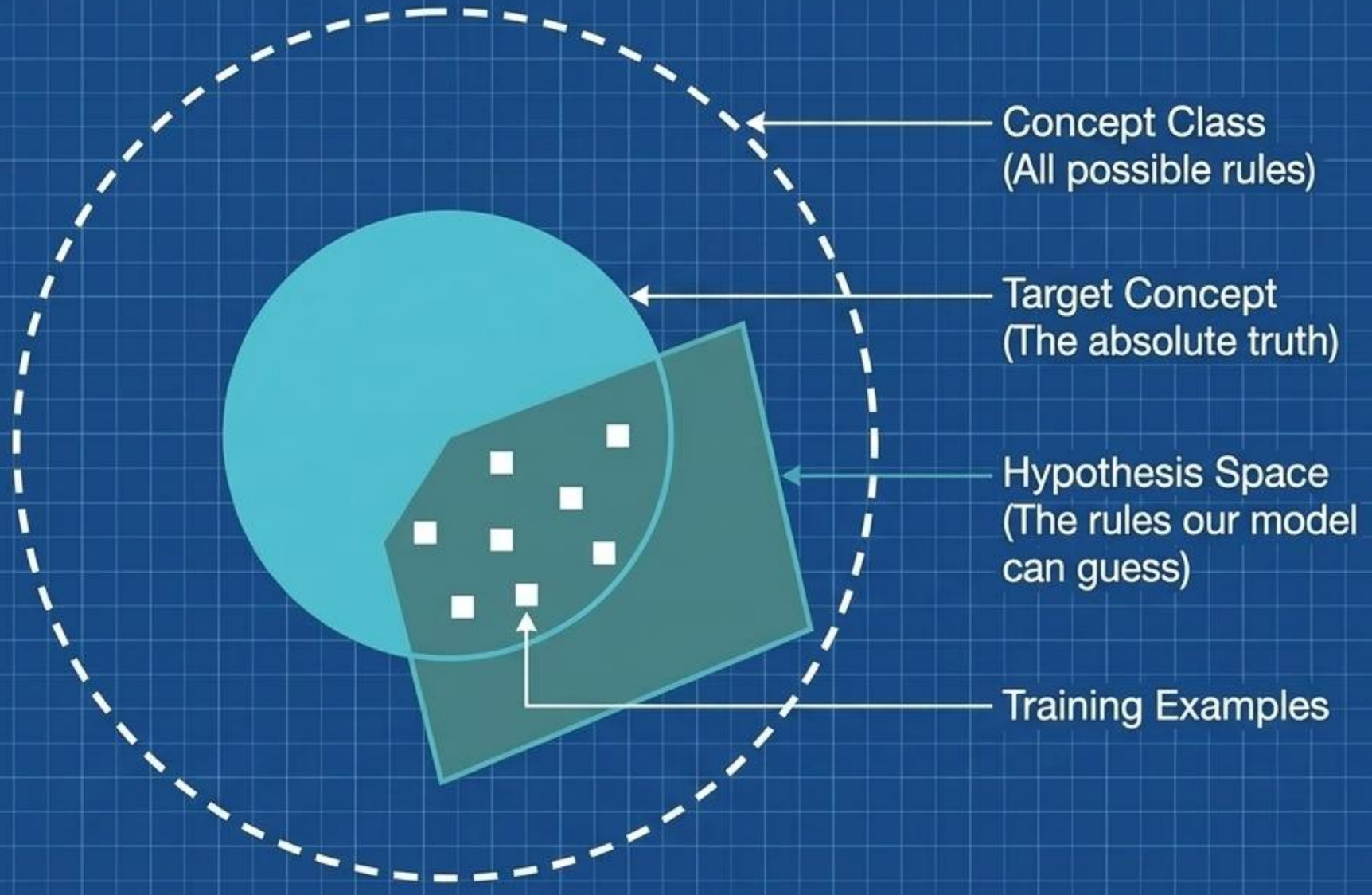
Correctness must be inferred from patterns.

The Generalization Problem:
We are blindly trusting a generated rule on a future we haven't seen.

Future Predictions

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Defining the Learning Universe



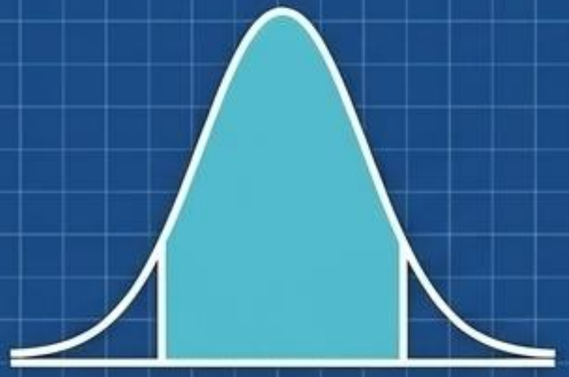
Student Pass/Fail Scenario

Training Examples
Historical student data (known outcomes).

Target Concept
The true universal law of what makes a student pass (unknown).

Hypothesis
Our model's best guess at that law.

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PROBABLY (THE RISK)

High Confidence.
We acknowledge that an unrepresentative training dataset could fool us, but the chance of that happening is extremely small.



APPROXIMATELY (THE TOLERANCE)

Small Acceptable Error.
We do not demand absolute perfection, but our model's error rate will stay within a strict, predefined limit.

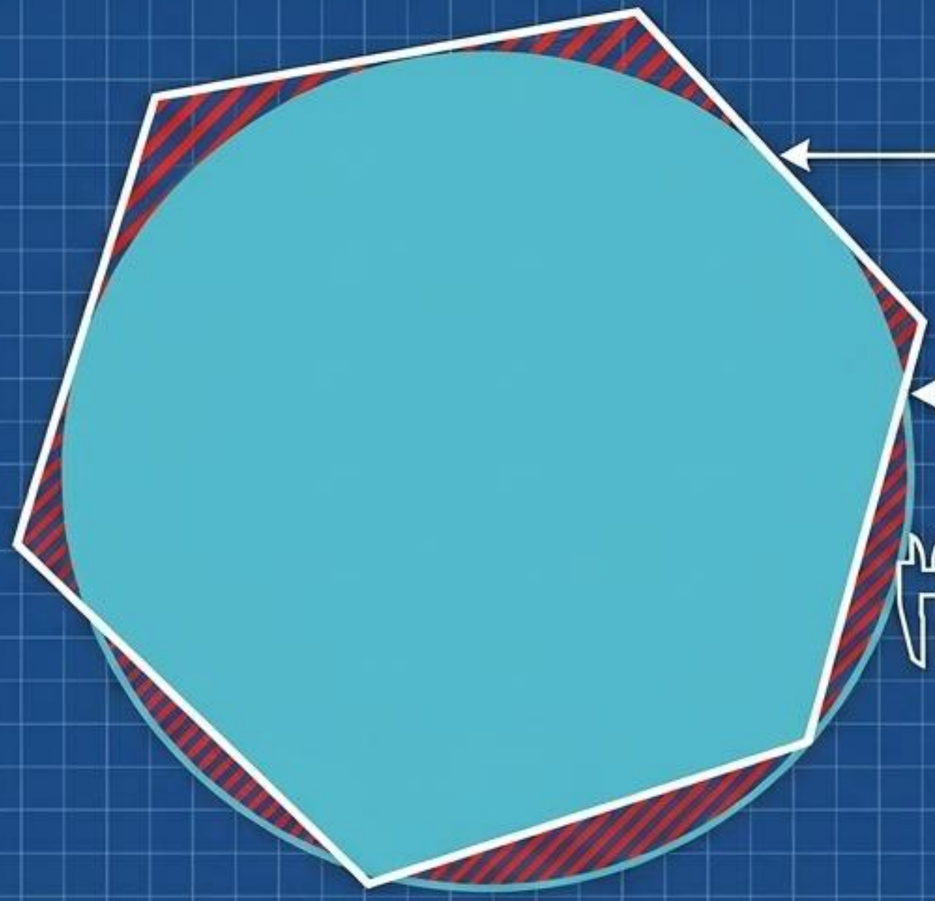


CORRECT (THE GOAL)

Future Performance.
The learned rule will classify the vast majority of future, unseen examples accurately.

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Understanding ε (Epsilon): The Allowed Error



Target Concept
(Absolute Truth)

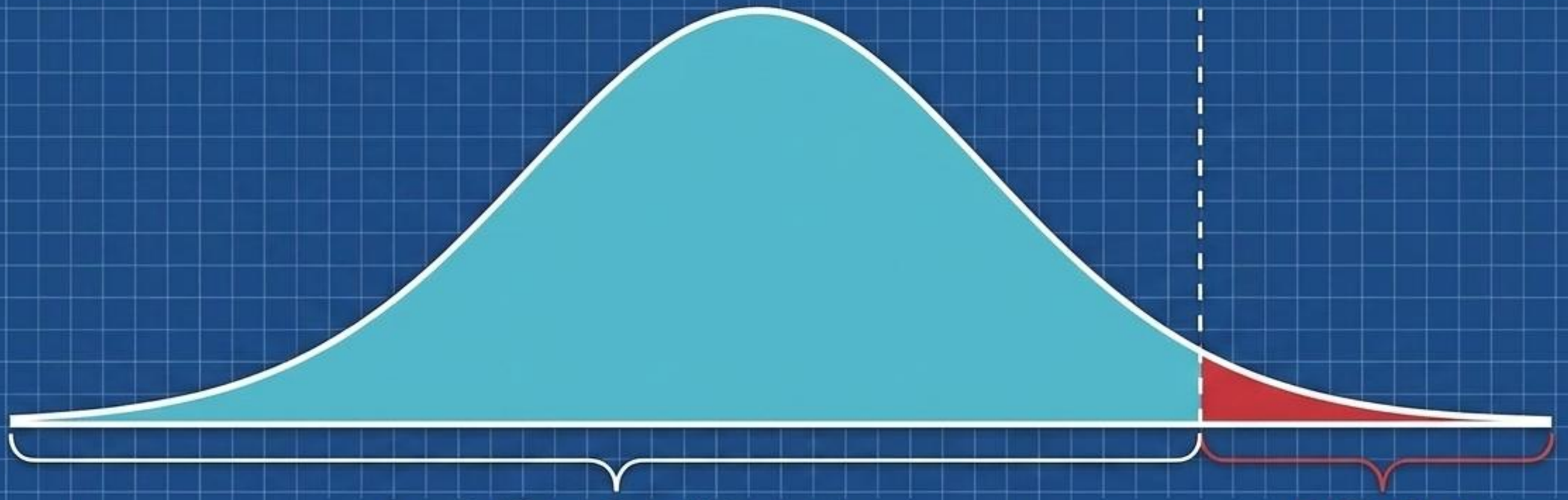
Hypothesis
(Model's Rule)

This gap is **Epsilon** (ε).
It represents the maximum acceptable error.

If $\varepsilon = 0.05$, and we process 100 new emails tomorrow, we accept that up to 5 may be classified incorrectly, while 95 will be perfectly correct.

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Understanding δ (Delta): The Confidence Level

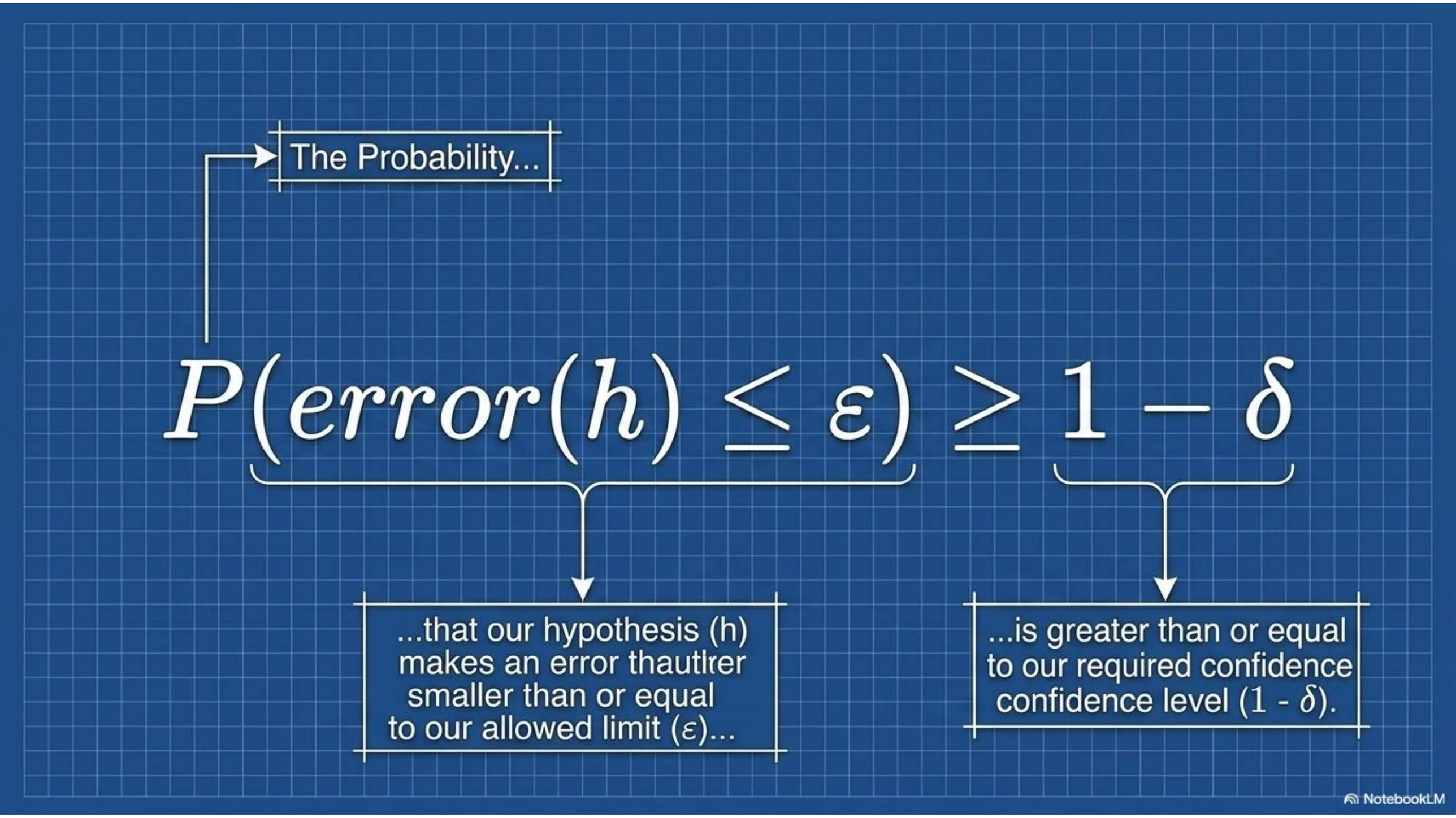


1 – δ (Our Confidence)
The probability that our training data accurately represents the real world.

δ (Delta)
The tiny probability of statistical bad luck—drawing a highly unrepresentative training dataset.

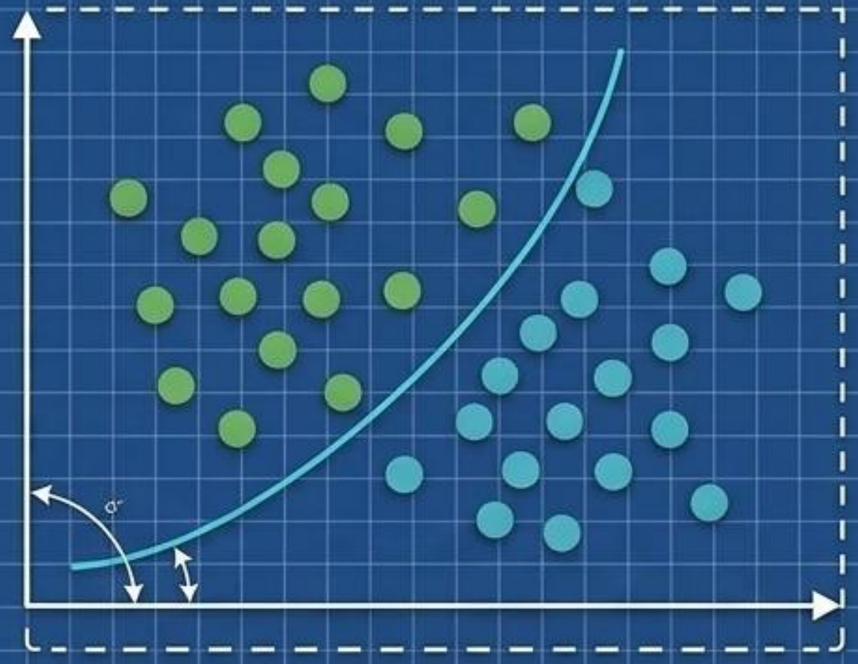
If $\delta = 0.05$, we are 95% confident that our model's error will not exceed our allowed tolerance (ε).

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The Probability...
$P(error(h) \leq \varepsilon) \geq 1 - \delta$
...that our hypothesis (h) makes an error thautlrer smaller than or equal to our allowed limit (ε)...
...is greater than or equal to our required confidence confidence level (1 - δ).

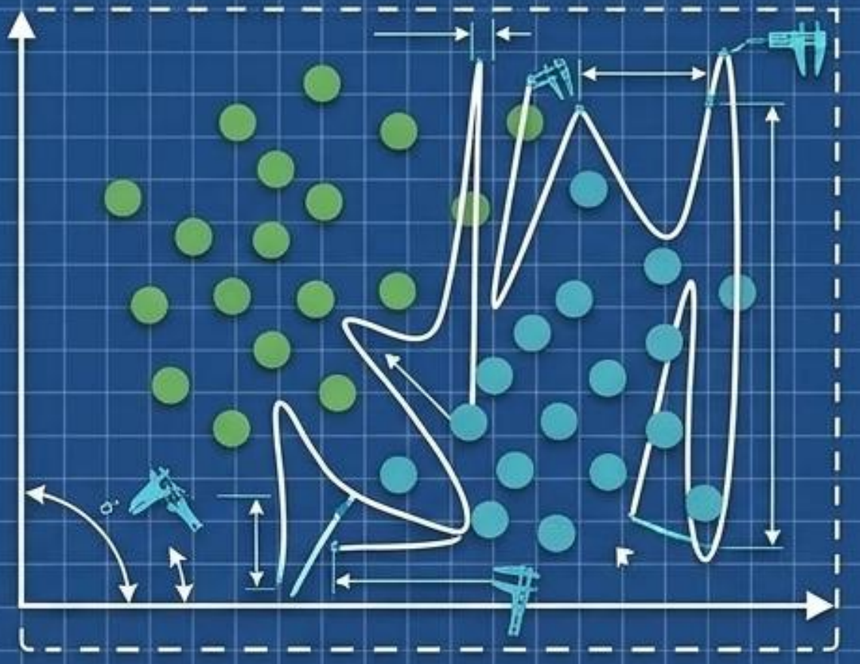
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Good Fit



Good Fit
Training Error: Low
Testing Error: Low
Generalization: Strong

Overfitted Model



Overfitted Model
Training Error: Zero
Testing Error: High
Generalization: Failed

The PAC Lesson: **Forcing ε to absolute zero on training data destroys the model's ability to handle future data.**

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Sample Complexity: How Much Data is Enough?

$$m \geq \frac{1}{\varepsilon})(\ln|H| + \ln(\frac{1}{\delta}))$$



m

The minimum number of training examples required.



$1/\varepsilon$

As allowed error shrinks, required data grows proportionally.



$|H|$

The size of the Hypothesis Space. Complex problems require exponentially more data.

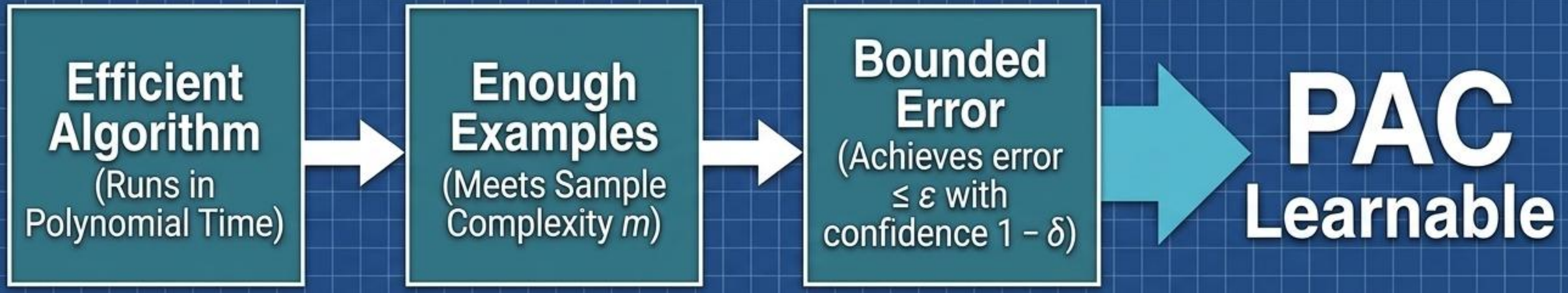


$1/\delta$

As required confidence increases, required data grows logarithmically.

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The PAC Learnability Engine



A concept is PAC Learnable if, and only if, an algorithm can efficiently process a manageable amount of data to produce a model that generalizes well.

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The Constraints of the Guarantee

PAC Assumptions (The Ideal World)	Practical Limitations (The Real World)
Data is drawn independently (I.I.D.).	Distributions shift over time (Tomorrow's spam looks different).
Future data uses the exact same probability distribution as training data.	Real datasets contain mislabeled data and human error.
Training labels are 100% correct (No noise).	Continuous variables create infinite hypothesis spaces.
The hypothesis space is finite.	Often serves as a theoretical bound rather than a practical engineering metric.

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Real-World Applications Governed by PAC Principles



SPAM & THREAT DETECTION



MEDICAL DIAGNOSIS MODELING



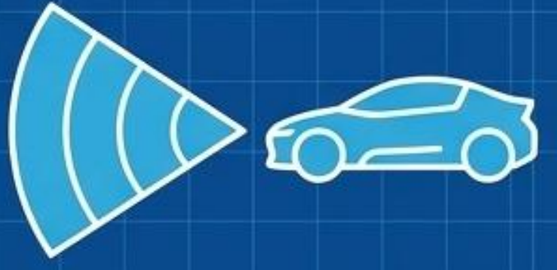
FACIAL RECOGNITION SYSTEMS



SPEECH-TO-TEXT PROCESSING



RECOMMENDATION ENGINES



AUTONOMOUS VEHICLE PERCEPTION

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The 5 Parameters of Learning (Summary)



Probably Approximately Correct (The framework for learning guarantees).

The absolute maximum error we will tolerate.

Our confidence level against statistical bad luck.

The universe of all possible rules our model can learn.

The mathematical minimum data required to make the guarantee.

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